

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	N.Adi Narayana	Reg. No.:	192525346
Section / Batch:		Date:	

Code of Conduct

I P. Chola Lakshmi Narasimha/192325021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Chola Lakshmi Narasimha

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, RuleBased Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

1. Morphological Analysis Pipeline for Document Indexing

Input Words

- connected
- connecting
- connection

Step 1: Morphological Decomposition

Word	Root	Suffix	Type
Connected	Connect	-ed	Inflectional
Connecting	Connect	-ing	Inflectional
Connection	Connect	-ion	Derivational

Step 2: Inflectional vs Derivational

- **Inflectional Morphology**
 - Does **not** change the word class.
 - Only changes tense, number, aspect, etc.
 - Example:
 - connect → connected
 - connect → connecting
- **Derivational Morphology**
 - Changes the meaning or grammatical category.
 - Example:
 - connect (verb)
 - connection (noun)

Step 3: Normalization

All words are normalized to the common base form:

connected → connect

connecting → connect

connection → connect

Morphological Parsing Output

Input word	Root	Affix	Classification	Normalized Output
Connected	Connect	-ed	Inflectional	Connect
Connecting	Connect	-ing	Inflectional	Connect
Connection	Connect	-ion	Derivational	Connect

Morphological Analysis Pipeline

Input Words



Tokenization



Identify Prefix/Suffix



Extract Root



Classify
(Inflectional / Derivational)



Normalize to Base Form



Store in Document Index

2. Morphological Parsing for Sentiment Analysis

Input Words

- unhappy
- happiness
- happily

Rule-Based Decomposition

Word 1: unhappy

un + happy

- Prefix = un-
- Root = happy
- Suffix = None

Meaning:

Negation of "happy"

Word 2: happiness

happy + ness

Root = happy

Suffix = -ness

Forms a noun.

Word 3: happily

happy + ly

Root = happy

Suffix = -ly

Forms an adverb.

Morphological Breakdown

Word	Prefix	Root	Suffix	Classification
Unhappy	Un-	Happy	-	Derivational
Happiness	-	Happy	-ness	Derivational
Happily	-	Happy	-ly	Derivational

Semantic Consistency

All words express the same emotional concept related to **happy**.

unhappy → happy

happiness → happy

happily → happy

Final Output

Input	Morphological breakdown	Normalized Root
Unhappy	Un+happy	Happy
Happiness	Happy+ness	Happy
Happily	Happy+ly	Happy

3. Stemming-Based Preprocessing Module

Input Words

- played
- player
- playing

Affix Stripping

played

played

remove -ed

Stem = play

player

player

remove -er

Stem = play

playing

playing

remove -ing

Stem = play

Classification

Word	Stem	Affix	Classification
Played	Play	-ed	Inflectional
Player	Play	-er	Derivational
Playing	Play	-ing	Inflectional

Consistent Root Extraction

played

player

playing

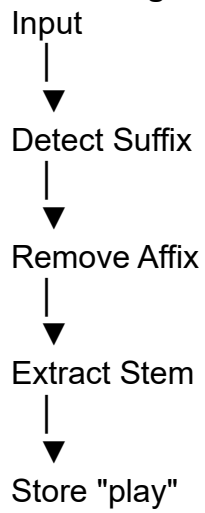
↓

play

Final Output

Input Word	Stem	Classification
played	play	Inflectional
player	play	Derivational
playing	play	Inflectional

Stemming Module



4. Finite-State Morphological Parsing

Input Words

- writes
- writing
- written

Finite-State Representation

writes

write

|

+ s

|

writes

writing

write

|

+ ing

|

writing

written

Irregular verb

write

|

Irregular transformation

|

written

State Transition Diagram

+-- s -----> writes

/

START → write

\

+-- ing -----> writing

\

+-- irregular ---> written

Morphological Breakdown

Word	Root	Affix	Pattern
writes	write	-s	Regular
writing	write	-ing	Regular
written	write	irregular (-en)	Irregular

Regular vs Irregular

Regular

- writes
- writing

Irregular

- written

Normalized Output

All words become

Write

Final Table

Input	State Transition	Morphological Breakdown	Normalized
writes	write → +s	write + s	write
writing	write → +ing	write + ing	write
written	write → irregular	write + en	write

5. Porter Stemmer-Based Preprocessing

Input Words

- relational
- relation
- relate

Word 1: relational

Step 1

relational

Porter Rule:

-ational → ate

relational

↓

relate

Step 2

relate

Remove trailing e

relat

Final stem:

relat

Word 2: relation

Step 1

relation

Remove

-ion

Intermediate

relat

Final stem

relat

Word 3: relate

Step 1

relate

Remove final **e**

relat

Final stem

relat

Step-by-Step Stemming

Word	Rule Applied	Intermediate Form	Final Stem
relational	-ational → ate	relate	relat
relation	Remove -ion	relat	relat
relate	Remove final e	relat	relat

Uniform Root Extraction

relational

relation

relate

↓

relat

Final Output

Input Word	Stemming Steps	Final Stem
relational	relational → relate → relat	relat
relation	relation → relat	relat
relate	relate → relat	relat

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance		Basic interpretation with minor omissions		Limited or unclear interpretation		No interpretation	
Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %		
1									
2									
3									
4									
5									